

# FATIMA MAMU

AI Product Engineer | Customer-Embedded Delivery | Generative AI & Agentic Systems

Boston, MA | Open to On-Site (Cambridge, MA), Remote & Relocation | 863-317-5742 |

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## PROFESSIONAL SUMMARY

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Harvard-trained AI Product Engineer who builds end-to-end at the intersection of design, engineering, and customer experience. I take ambiguous enterprise problems through discovery, prototyping, production deployment, and the messy iteration that happens once real users get their hands on something. My work spans agentic and LLM-powered systems, real-time AI products, and customer-embedded delivery, including production AI agents shipped into live clinical environments and full-stack prototypes designed for technical and non-technical decision-makers. I move fluidly between writing Python and TypeScript, designing flows in Figma, demoing live to executives, and debugging the long tail of issues that show up only in real deployments. Equally comfortable as a builder, a designer, a technical translator, and the person sitting in front of a customer when something needs to be solved fast.

## SIGNATURE STRENGTHS

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- **Designer–Engineer:** Harvard MDE foundation in cross-disciplinary product building, fluent in Figma, design systems, and interaction design as well as Python, TypeScript, and modern AI/LLM stacks.
- **Generative AI & Agentic Systems:** Build with LLMs daily, agentic workflows, multi-step reasoning, retrieval-augmented generation (RAG), structured outputs, evaluation harnesses, and prompt engineering as a production discipline.
- **Customer-Embedded Delivery:** Deployed on-site in live clinical environments and at client sites across multiple industries; trained as a forward-deployed builder who turns observed friction into shipped fixes.
- **Technical Translation & Demos:** Run live product demos for technical and executive audiences; can move from a CTO conversation to a clinician conversation to a stakeholder readout in the same afternoon.
- **Speed Without Sacrificing Craft:** Built for startup velocity, ship fast, iterate publicly, and notice the details others overlook.

## TECHNICAL & DESIGN SKILLS

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**Languages & Engineering:** Python, TypeScript / JavaScript, SQL; comfortable reading and modifying production codebases; debugging logs, traces, and distributed-system behavior.

**Generative AI & LLMs:** LLM application development, agentic and multi-step workflows, RAG, prompt engineering, function calling / tool use, structured outputs, evaluation and benchmarking, human-in-the-loop pipelines, fine-tuning fundamentals (post-training).

**AI Tools & Platforms:** Daily fluency with frontier LLMs (OpenAI, Anthropic, Gemini Live, Gemini 2.5 Flash); AI-assisted coding tools; rapid integration of new models and modes the day they ship, AI-first by default.

**Product Design & UI Craft:** Figma (components, variants, auto-layout), wireframes and high-fidelity UI, interaction design, design systems, edge-case and error-state thinking, engineer-ready specs and handoffs, accessibility and design-for-clarity.

**APIs, Backend & Infrastructure:** FastAPI, REST and streaming APIs, webhook integrations, PostgreSQL, Supabase, pgvector (vector search); Google Cloud (Cloud Run, Cloud SQL, Secret Manager); Docker, containerized deployments, CI/CD.

**Customer-Facing Engineering:** Technical discovery, proof-of-concept (POC) design and execution, live demos, architecture reviews, on-site enterprise customer engagement, escalation ownership, root-cause analysis, customer-feedback-to-product-improvement loops.

**Real-Time, Voice & Interactive Systems:** Real-time speech (STT/TTS), streaming audio, low-latency pipelines, voice agents, sensor-driven interactive experiences (Unity, TouchDesigner).

## PROFESSIONAL EXPERIENCE

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### **Founding Engineer** | The One Care, Boston, MA

**2026 – Present**

- Architected and shipped a production AI agent — combining LLM reasoning with structured retrieval over pgvector / PostgreSQL — to deliver multi-turn, context-aware interactions; achieved ~25% improvement in response relevance across ~1,500 evaluation queries.
- Designed and built an end-to-end pipeline that converts unstructured input into validated structured records in real time, with schema-level validation, latency budgets tuned for live interaction, and graceful handling of ambiguous, accented, and incomplete inputs.
- Embedded directly with end users as the primary technical contact — running discovery sessions, observing live usage, triaging issues, and translating field signals into prioritized product and system improvements; functioned as the bridge between users and engineering.
- Built an automated ingestion and classification pipeline using Gemini 2.5 Flash with a human-in-the-loop review layer to scale curation of domain content into a high-quality, retrieval-ready knowledge index.
- Stood up the cloud footprint on Google Cloud (Cloud Run, Cloud SQL, Secret Manager) and built a FastAPI-based evaluation harness with structured test cases for ambiguous, incomplete, and conflicting inputs — used as the gate before any production deploy.
- Owned cost / latency / accuracy trade-offs across the stack, with structured logging and metrics that let model and prompt changes be evaluated against real production traffic, not just offline benchmarks.

### **AI Product Engineer** | Lexi AI, Boston, MA

**2025 – 2026**

- Shipped customer-facing AI features deployed across 4 enterprise facilities and 7,000+ live conversations, improving end-user efficiency by ~40% through real-time intelligence, in-product visual cues, and workflow-integrated controls.
- Designed, built, and deployed an AI intake agent on Gemini Live, handling end-to-end real-time interaction: low-latency understanding, conversational reasoning, structured data extraction, and API-based routing into downstream provider systems (JotForm). Built specifically for low-digital-literacy users on standard telephony.
- Operated as the on-site technical lead in live customer environments: shadowing end users, diagnosing breakdowns in real workflows, owning escalations end-to-end, and turning observations into reliability, UX, and prompt improvements shipped within the same sprint.
- Owned the technical sales narrative for prospective facilities — running live demos, designing POCs, answering deep technical questions on accuracy, latency, integration patterns, and data handling, and producing clear written materials for technical and non-technical stakeholders.
- Built internal evaluation tooling to track model quality, agent task-completion rates, and end-to-end latency; surfaced regressions before they reached customers and translated patterns across support cases into product improvements.

### **Software Development Intern, R&D** | Walt Disney Imagineering, Glendale, CA

**2024**

- Built internal tools for the R&D team, including a synthetic dataset generation pipeline for 6-DoF pose estimation and an AR pre-production staging environment, reducing physical setup time and improving spatial accuracy before on-site installation.
- Designed and developed real-time, sensor-driven interactive experiences in Unity, Maya, and TouchDesigner — including a guest-facing water simulation for the Moana: Journey of Water exhibit (D23) that translated full-body motion tracking into responsive visual effects with sub-frame latency requirements.
- Integrated robotics, real-time animation, and physical staging for a KUKA robot-based illusion, synchronizing TouchDesigner animations with LED outputs; presented the working system to Bob Iger, securing approval for project continuation.
- Worked across hardware, software, design, and creative stakeholders — practicing the same translation skill required in customer-embedded technical roles: making complex real-time systems legible to non-engineers.

### **Engineering Associate (Prototyper)** | The Business People Limited, Nigeria

**2019 – 2023**

- Acted as primary technical lead on enterprise client engagements across agriculture, aviation, and healthcare — owning requirements discovery, system design, stakeholder communication, and coordination with external engineering teams; equivalent to a hands-on solutions consultant / forward-deployed builder role.

- Designed and built full-stack prototypes (Next.js, TypeScript, PostgreSQL) covering UI, data modeling, API design, and end-to-end user flows; used these prototypes to validate product direction with executive stakeholders and de-risk full-scale builds before commitment.
- Translated ambiguous operational challenges from non-technical clients into concrete technical specifications, evaluation criteria, and demo-ready POCs — directly analogous to leading technical discovery and POC design in an enterprise sales cycle.
- Presented technical solutions to executive and operational stakeholders, defending architecture choices on cost, scalability, and reliability; iterated based on feedback from real users.

## Founder & Creative Director | Edible Art Studio, Nigeria

2013 – 2023

- Founded and ran a bespoke design studio, leading a 15-person team and delivering thousands of commissioned client projects over a decade — built deep instincts in visual craft, customer relationships, and operational systems-thinking that translate directly into modern product work.
- Designed and implemented operational systems for client intake, custom order management, and supplier coordination, enabling consistent, scalable delivery — the same systems-design instinct I now bring to building agentic AI workflows.

## EDUCATION

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| <b>Harvard University</b> , Cambridge, MA — Master in Design Engineering (MDE), Harvard SEAS & GSD     | 2023 – 2025 |
| <i>MDE Fellowship recipient (merit-based); Leadership Award for cross-disciplinary collaboration.</i>  |             |
| <b>University College London (UCL)</b> , London, UK — Diploma in Computer Graphics, Vision and Imaging | 2015 – 2016 |
| <b>Brunel University London</b> , London, UK — BEng, Electrical and Electronics Engineering            | 2010 – 2013 |

## HONORS, AWARDS & CERTIFICATIONS

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- **Harvard University** — Leadership Award (2025); MDE Fellowship (2024).
  - **Microsoft** Full-Stack Developer Professional Certificate.
  - **IBM** AI Product Manager Professional Certificate.
  - **Specialized coursework:** Machine Learning in Production; Fine-Tuning & Reinforcement Learning for LLMs (Intro to Post-Training); A2A — The Agent2Agent Protocol; Leveraging Data in Iterative Product Design.

## SELECTED PROJECTS

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- **Production AI Agent for Caregiving (The One Care)** — End-to-end LLM agent with pgvector RAG, schema-validated structured outputs, and a FastAPI evaluation harness; demonstrates ownership of an agentic system from architecture to live deployment.
  - **Phone-Based AI Intake Agent (Lexi AI)** — Telephony-integrated voice agent on Gemini Live for low-digital-literacy users, with structured data extraction and routing to downstream enterprise systems.
  - **Evaluation Harness for AI Agents** — FastAPI-based test framework with curated edge-case datasets (ambiguous, incomplete, conflicting inputs), used to gate model and prompt changes pre-deployment.
  - **Real-Time Interactive Experiences (Disney Imagineering R&D)** — Unity + TouchDesigner systems including the Moana: Journey of Water D23 exhibit; design and engineering combined under sub-frame latency constraints.
  - **Enterprise Prototype Portfolio (Cross-Industry)** — Full-stack prototypes (Next.js, TypeScript, PostgreSQL) shipped as decision-making artifacts for clients in agriculture, aviation, and healthcare.

## HOW I OPERATE

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- Builder first. I'd rather show a working POC than write a long spec.
  - AI-first by default. New models and modes are integrated into my daily workflow the day they ship; AI tools are how I move at startup speed.
  - Comfortable in the messy middle. A decade of customer-facing work before engineering taught me how to sit with ambiguity, ask the right questions, and translate between technical depth, product thinking, and the people who actually use what we build.
  - Notice what others overlook. The detail in the error state, the corner case in the flow, the off-by-one in the demo — that level of care is non-negotiable for me.